PRESTIGE INSTITUE OF ENGINEERING, MANAGEMENT AND RESEARCH
DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

DESIGN DREAMER
DESIGN YOUR DREAM SPACE WITH STATE-OF-THE-ART GENERATIVE AI

DesignDreamer: Design your Dream Space with state-of-the-art Generative AI

Dr. Dipti Chahuhan[1], Param Dhingana[2], Vinamra Khandelwal[3], Sudeep Ganguly[4]

[1] HOD & Associate Professor, Department of Artificial Intelligence and Data Science, PIEMR, Indore

[2], [3], [4] Student, B. Tech, Department of Artificial Intelligence and Data Science, PIEMR, Indore

Problem Statement

Traditional interior design faces issues like limited personalization, time-consuming processes, and visualization difficulties.

DesignDreamer solves these challenges using Generative AI and advanced technologies.

Abstract

DesignDreamer represents a futuristic vision in the realm of Generative AI, harnessing advanced technologies to elevate interior design into a personalized and visually stunning experience. This project delves into the intricate fusion of user-uploaded room images, cutting-edge image generation and recreation models, and design preferences to create tailored and aesthetically pleasing spaces. By seamlessly integrating style preferences and room types, DesignDreamer empowers users to effortlessly visualize and co-create their dream environments.

Introduction

In the era of advanced artificial intelligence, DesignDreamer emerges as a pioneering project that caters to the innate human desire for personalized and visually appealing living spaces. Traditional interior design processes often lack individualization and can be time-consuming. DesignDreamer addresses these challenges by integrating Generative AI, enabling users to actively participate in the creation of their dream spaces. With a simple image upload, users can navigate through various design styles and room types, receiving a tailored, AI-generated representation of their chosen preferences.

Keywords

Generative AI, Interior Design, Customization, Image Processing, Style Transfer, Python, Streamlit, Replicate, API, Personalization, Visualization

Technology

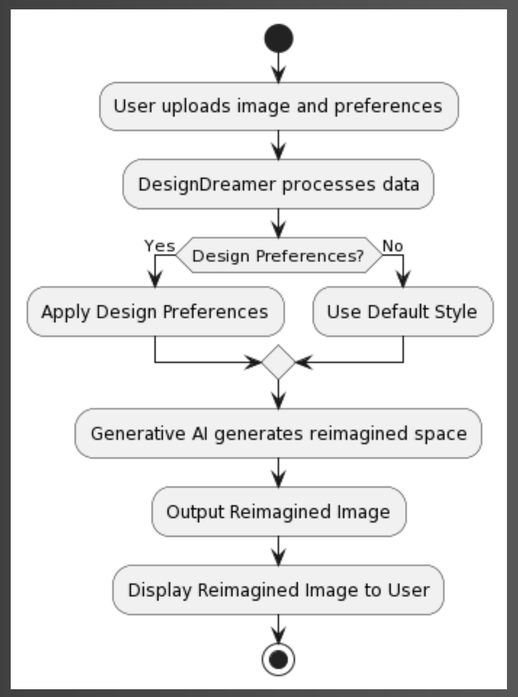
Replicate

python™

jagilley/ controlnet-hough
Modify images using M-LSD line detection

Streamlit

System Diagram



Applications

- Personal Interior Design
- Real Estate
- Hospitality Industry
- Furniture Retail
- Architectural Visualization